



DPURacingWheel

Provides an interface to joysticks and steering wheels. If available, force feedback is supported.

1. Simple Force Feedback Input:

Force Force feedback to the driver. The range is between -1 and +1. The direction of the force depends on the driver implementation.

2. Advanced Force Feedback Inputs:

For each of the four force-feedback subsystems *Spring*, *Damper*, *Friction*, and *Inertia* (details: see below), the following variables below *SystemName*. exist:

<i>Offset</i>	Center of axis (where no force is applied) [-1..1]
<i>DeadZone</i>	Dead zone around the offset; no force is applied within the range [Offset-DeadZone .. Offset+DeadZone] [0..1]
<i>NegativeCoefficient</i>	Linear slope of the force, if the axis value is less than <i>Offset</i> [-1..1]
<i>PositiveCoefficient</i>	Linear slope of the force, if the axis value is greater than <i>Offset</i> [-1..1]
<i>NegativeSaturation</i>	Saturation value of force, if the axis value is less than <i>Offset</i> [0..1]
<i>PositiveSaturation</i>	Saturation value of force, if the axis value is greater than <i>Offset</i> [0..1]

3. Static parameters:

<i>Device</i>	Name of the joystick to open. If left empty (the default), the first connected joystick is used. The names of all connected joysticks are logged when the DPU is loaded.
<i>SetDefault</i>	Some joysticks start with non-centered axes, but do not report the axis location before it is moved. In such cases, set <i>SetDefault</i> = <i>true</i> and enter the correct startup values under the <i>Default_x</i> parameters. Default: false, which works for most joysticks.
<i>Default_x, Default_y, ... Default_rotz</i>	Startup values for non-centered axes. Only used if <i>SetDefault=true</i> was entered. The range is from -1 to +1. Default: 0
<i>AlternateForces</i>	If set to false (the default), all subsystems are updated in every simulation step. If set to true, then only one of the advanced force feedback subsystems is set per simulation step. Use this setting if the device driver takes a long time to update the force feedback settings.



4. Outputs:

<i>x</i>	Value of the steering wheel's X axis (normally corresponds to left/right movement, between -1 (left) and +1 (right))
<i>y</i>	Value of the steering wheel's Y axis (normally corresponds to back/forward movement, between -1 (back) and +1 (forward))
<i>z</i>	Value of the steering wheel's Z axis (often corresponds to the „throttle“ control, between -1 (low) and +1 (high))
<i>rotx</i>	Value of the steering wheel's rotation around the X axis (range: -1 to 1)
<i>roty</i>	Value of the steering wheel's rotation around the Y axis (range: -1 to 1)
<i>rotz</i>	Value of the steering wheel's rotation around the Z axis (range: -1 to 1)
<i>Button[1...20]</i>	State of the buttons (0: unpressed, 1: pressed)
<i>Slider[1...4]X</i>	Value of the sliders in horizontal direction (range is not standardized)
<i>Slider[1...4]Y</i>	Value of the sliders in vertical direction (range is not standardized)
<i>Pov[1...4]</i>	Position of the coolie hat („point of view hats“). The encoding of the variable is not standardized.
<i>Pov[1...4]N</i>	Boolean output, which is set to <i>true</i> when this coolie hat points north.
<i>Pov[1...4]NW</i>	Boolean output, which is set to <i>true</i> when this coolie hat points north-west.
<i>Pov[1...4]W</i>	Boolean output, which is set to <i>true</i> when this coolie hat points west.
<i>Pov[1...4]SW</i>	Boolean output, which is set to <i>true</i> when this coolie hat points south-west.
<i>Pov[1...4]S</i>	Boolean output, which is set to <i>true</i> when this coolie hat points south.
<i>Pov[1...4]SO</i>	Boolean output, which is set to <i>true</i> when this coolie hat points south-east.
<i>Pov[1...4]O</i>	Boolean output, which is set to <i>true</i> when this coolie hat points east.
<i>Pov[1...4]NO</i>	Boolean output, which is set to <i>true</i> when this coolie hat points north-east.

5. Functional principle:

The DPURacingWheel represents an interface to any joystick or steering wheel that is supported by the operating system.

On Windows, all devices compatible to Microsoft DirectX 8.0 are supported. Force feedback is supported by the DPU if the device supports it.



On Linux, all joysticks and steering wheels supported by the Linux kernel can be used for input. However, please note that force feedback is currently supported only for very few steering wheels (mostly the Logitech G series), and even for them, only *simple force feedback* is supported.

The axis range is standardized between -1 and +1.

The DPU only creates the output variables that are reported by the device. However, some devices report more axes or buttons than they actually have. The values of these axes or buttons are set to zero.

6. Force Feedback:

There are two force feedback systems which can be used with DPURacingWheel: the *simple* and the *advanced force feedback system*.

Both systems can be used at the same time, if desired.

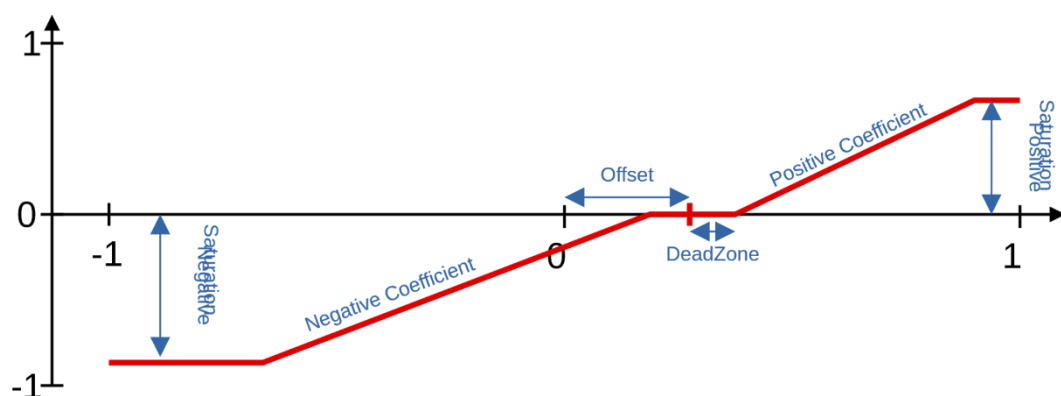
The *simple force feedback system* consists of the input variable *Force*. The range of this variable is between -1 (maximum force towards the left) and +1 (maximum force towards the right).

The *advanced force feedback system* consists of four modules:

- *Spring* simulates a spring which centers the steering wheel at a desired position. The force is proportional to the angular distance from the desired position.
- *Friction* makes it harder to move the wheel, independent of speed and direction.
- The *Damper* makes it harder to move the wheel, proportional to the speed of movement.
- The *Inertia* system generates can be used (using negative coefficients) to implement a “slugish” wheel which keeps moving for a little longer even after it has been released by the driver. The force is proportional to the acceleration of the wheel.

For each system, a piecewise linear equation is defined using six input parameters. It maps the input value (position, speed or acceleration, normalized to the range -1..1) to an output force (also in normalized range -1..1).

The equation is defined as follows:





The meaning of each parameter is in detail:

<i>Offset</i>	Center of axis (where no force is applied). Example: For the <i>Spring</i>, it's the position to which the wheel centers.
<i>DeadZone</i>	Dead zone around the offset; no force is applied within the range $[Offset-DeadZone .. Offset+DeadZone]$ Example: For the <i>Spring</i>, this is an area around the <i>Offset</i> position where the wheel can be moved without any resistance.
<i>NegativeCoefficient</i>	Linear slope of the force, if the axis value is less than <i>Offset</i>. Can be in range -1..1, where negative values reverse the direction of the effect. Example: For the <i>Spring</i>, this is the strength of the spring while moving to the left of the <i>Offset</i> position.
<i>PositiveCoefficient</i>	Linear slope of the force, if the axis value is greater than <i>Offset</i>. Can be in range -1..1, where negative values reverse the direction of the effect. Example: For the <i>Spring</i>, this is the strength of the spring while moving to the right of the <i>Offset</i> position.
<i>NegativeSaturation</i>	Saturation value of force, if the axis value is less than <i>Offset</i>. It is always given as a positive value (0..1). Example: For the <i>Spring</i>, this is the maximum strength of the spring on the left side of the <i>Offset</i> position.
<i>PositiveSaturation</i>	Saturation value of force, if the axis value is greater than <i>Offset</i>. It is always given as a positive value (0..1). Example: For the <i>Spring</i>, this is the maximum strength of the spring on the right side of the <i>Offset</i> position.